Kang Hyun(강현)

Seoul National University, Department of Industrial Engineering
mystic030115@snu.ac.kr — Seoul, South Korea

Education

Seoul National University Seoul, South Korea
B.S. in Industrial Engineering; Double Major in Artificial Intelligence Mar. 2022 – Expected Aug. 2029

- GPA: 3.79/4.30
- Relevant coursework: Optimization, Probability and Statistics, Computer Programming, Engineering Economy, Management Science, Real Analysis

Seoul Science High School Seoul, South Korea
High School Diploma Graduated 2022

Research Interests

Operations Research; Optimization; Stochastic Modeling; Robust and Data-Driven Decision Making; Supply Chain and Logistics Analytics; AI for Operations.

Industry Experience

Machine Learning & Software Engineer Intern, Waddle Corp. Seoul, South Korea
React/Next.js, NestJS, TypeORM, MariaDB, AWS, Superset Nov. 2025 – Mar. 2026

- Developed a product-information extraction pipeline using OCR and vision-language models to support automated product data processing.
- Implemented S3 upload logic and database insert/update workflows for production-facing services.
- Built analytics dashboards using Superset, Athena, and Presto to monitor user behavior and product metrics.
- Developed an LLM-based product recommendation system by fine-tuning a language model on product metadata and user interaction data to improve personalized item matching.

Research Experience

Undergraduate Research Student, SNU IE DSBA Lab Seoul, South Korea
Department of Industrial Engineering, Seoul National University Mar. 2026 – Present

- Participating in research seminars and VLM research meetings on vision-language models and data-driven business analytics.
- Reviewing recent literature on multimodal learning, product information extraction, and applied machine learning systems.
- Studying practical applications of VLMs for product understanding, recommendation, and e-commerce analytics.

Undergraduate Research Student, SNU Math OptA Lab Seoul, South Korea
Department of Mathematical Sciences, Seoul National University May 2026 – Present

- Conducting research on end-to-end LLM-based optimization solution pipelines.
- Studying multi-objective optimization algorithms and their applications to automated decision-making systems.
- Reviewing optimization methods for integrating mathematical modeling, algorithm selection, and solution interpretation.

Honors and Awards

- Silver Award, Korean Mathematical Olympiad Second Round 2018
- Completed KMO Winter School 2018–2019
- Top Award and three Excellence Awards in student research projects, Seoul Science High School

Technical Skills

Programming: Python, Java, C++, JavaScript/TypeScript, SQL
ML/Data: PyTorch, NumPy, Pandas, scikit-learn
Web/Backend: React, Next.js, NestJS, TypeORM, MariaDB
Cloud/Data: AWS S3, RDS, Elastic Beanstalk, Superset, Athena, Presto
Tools: Git, GitHub, Postman, Excel

Military Service

Republic of Korea Army South Korea
Sergeant, Instructor at Army NCO School 2024 – 2025